

Optical Fiber Cables and Splices

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Abstract—Requirements for optical fiber cables and splices differ depending on the area of application within the telecommunication network. This paper reviews optical fiber properties along with cable and splice designs, primarily for interoffice trunk applications, and gives Bell System results for this major area of optical fiber technology. These extremely favorable results indicate that optical fibers will play a major role in all segments of the telecommunication network: long-haul, trunk, and distribution.

I. INTRODUCTION

WITHIN the past five years, optical fiber communication systems have been developed and installed at an extremely fast rate. Most initial applications of optical fibers have been in the interoffice trunk segment of the telephone network where small high-capacity optical cables conserve valuable underground metropolitan duct space. Other smaller scale but highly significant applications are aimed at providing wide bandwidth signals directly to customers. These applications are more economically speculative, and requirements for low-cost cables and splices are significantly different compared to trunk applications. Optical fiber systems for the remaining area of the network, long-haul terrestrial and intercontinental (undersea) transmission, are now being developed. These high bit rate long repeater span systems will typically require yet another cable and splicing technology relating to single-mode fibers. Although work is underway to design optical fiber systems for all areas of the telecommunication network, by far the most mature application is the interoffice trunk.

This paper begins with a brief description of the relevant optical and mechanical properties of fibers and continues with various approaches to cable and splicing design with emphasis on interoffice trunk requirements. A brief summary of the results of the Bell System's commercial trunk applications is given; and the paper concludes with a description of cable and splice design requirements for the long-haul and distribution portions of the network.

II. OPTICAL AND MECHANICAL PROPERTIES OF FIBERS

The small size, low loss, and wide bandwidth of optical fibers [1] are attractive properties; however, cable and connection designs must retain these desirable properties during field installation and afterwards for many years. The statistical time-dependent physical strength of glass fibers, the small elongation to fracture compared to metallic conductors, along with bending and microbending properties make optical cable design [2]–[4] important and difficult.

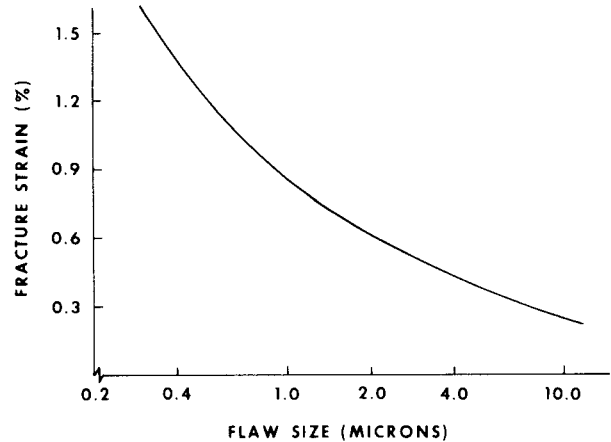


Fig. 1. Fracture strain versus flaw size in high silica fibers.

A. Physical Strength of Glass Fibers

It is well known that glass fibers elongate nearly elastically under tensile loading until brittle failure occurs. Cracks or flaws on the fiber surface cause failure at small loads compared to the theoretical strength of glass. Fig. 1 shows that, to achieve a tensile strain of one percent, the maximum flaw size must be less than approximately 1 μm [5]. Since some flaws of this size are likely to occur in a manufacturing environment, proof testing is employed to guarantee minimum short-term strength. Proof testing involves subjecting every segment of the fiber to a fixed controlled elongation typically between 0.5 and 1.0 percent.

B. Static Fatigue

The tensile strength of glass fibers degrades under strain in the presence of moisture, and, as a result, failure can eventually occur at strain levels significantly below those shown in Fig. 1. This failure is caused by time-dependent crack growth; and several models are available to describe this phenomenon. These models agree with static fatigue data out to a year or so; however, data are not available to indicate which, if any, of the models is accurate for periods of many years. To illustrate the sensitivity of static fatigue to initial strain capability, one model [5] predicts that a fiber with an initial strain capability of 0.33 percent, when exposed to 0.2 percent strain at 2 percent relative humidity will break in a few hours; whereas a fiber with 0.47 percent initial strain capability under the same conditions will survive 100 years.

The practicality of using fiber proof testing to insure a minimum in-cable strain capability is demonstrated [3] in Fig. 2. At 151 m cable containing 137 fibers, proof tested at 0.33 percent strain, was tensile tested by gradually increasing cable strain to 0.9 percent and maintaining that

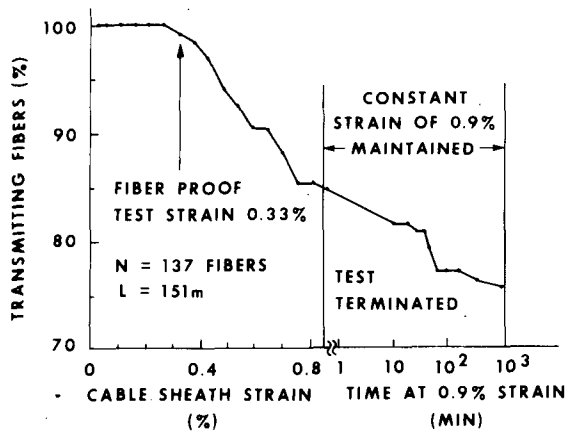


Fig. 2. Proof testing effectiveness and static fatigue behavior.

level for 10^3 min. No fibers in the cable broke until the cable strain was greater than the proof-test strain. This figure also illustrates static fatigue behavior in that fibers continued to break when a constant 0.9 percent strain was maintained.

C. Bending and Microbending

Bends smaller than a 3 cm radius should be avoided from static fatigue considerations since the surface of the fiber on the outside of the bend is under significant tensile strain. Pure bending losses for ~ 4 cm radius of curvature or greater are negligible over short distances but can accumulate [2] to ~ 1 dB/km, if cable designs cause this radius of curvature over a 1 km length. Microbending, small amplitude, and short period random bending of the fiber axis, can cause large added loss [6]. For typical multimode graded-index fibers, the approximate period at which microbending losses occur is on the order of 1 mm. The fiber coating, particularly coating smoothness and softness, are important design considerations to minimize microbending. Hard coatings tend to transmit any applied forces directly to the fiber; however, if these forces are controlled by the cable design, and if the coating is smooth, microbending losses will be small. Soft coatings reduce the effects of applied forces and are often used as a buffer layer with a hard outer coating to form a dual coating.

D. Fiber Size and Count

The last and perhaps most important physical property of fibers is their small size. Most telecommunication fibers are $125\ \mu\text{m}$ outer diameter, and although small size is a distinct space advantage compared to copper media, fibers of this size are difficult to handle individually, especially in an outside plant manhole environment. For intercity trunk applications, it is not unusual to require 100 or more fibers per cable. It is difficult to imagine the level of tedium involved in handling 200-300 (two cables to be spliced) barely visible thread-like fibers and performing the many operations required in individual fiber splicing and organization of the completed cable splice. Mass splicing techniques can be used to advantage, if the number of fibers is greater than perhaps 20, and if the cable has been designed for mass splicing.

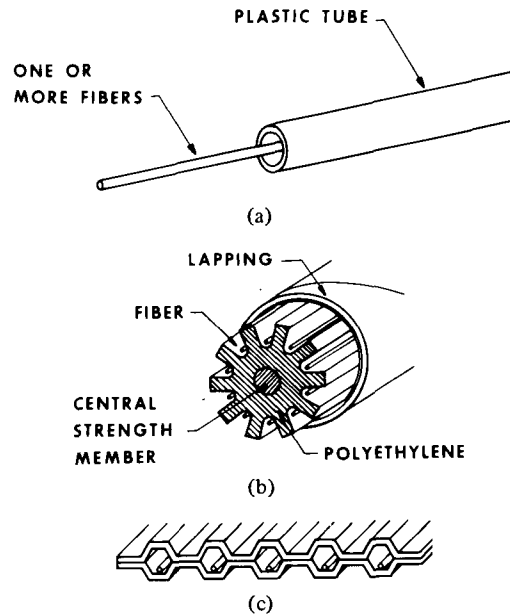


Fig. 3. Loose core structures. (a) Loose tube. (b) Loose cylindrical structure. (c) Loose ribbon.

III. CABLE STRUCTURES

Almost every fiber cable manufacturer has a different optical cable design. Fundamental elements of an optical cable include: 1) loose or tight core structures for supporting individual or groups of fibers, 2) strength members, and 3) a cable sheath containing organized core structures. Cables for trunk applications may require some open space for pressurization and usually require high tensile strength for pulling long cables in old curvy intercity ducts.

A. Core Structures

Both tight and loose core structures have been successfully manufactured and used commercially with low added loss during cabling. Loose structures include those in Fig. 3 in which the fiber is loosely held in an oversized tube [7], a grooved cylindrical structure [8], or a loose ribbon [9]. The typical goal for these designs is to ensure that the fiber length and core supporting structure are of equal length. Since there is little frictional coupling in these structures, the condition of equal length is somewhat difficult to achieve during manufacture; however, many manufacturers have developed techniques that work. Added loss due to cabling can be near zero if the equal length condition is met. Some manufacturers design in a given degree of "looseness" (excess area around the fiber) to allow greater variation in temperature or to store a small amount of excess fiber length in the loose spaces to allow large elongation of the cables. Tight structures often have extra layers of soft buffering material extruded around the primary coated fiber with a hard plastic outer coating. The adhesive ribbon [10] is one of the tightest structures; however, added loss in this structure can be kept negligibly low with either soft or smooth hard fiber coatings. Tight structures can be compact and have the advantage that

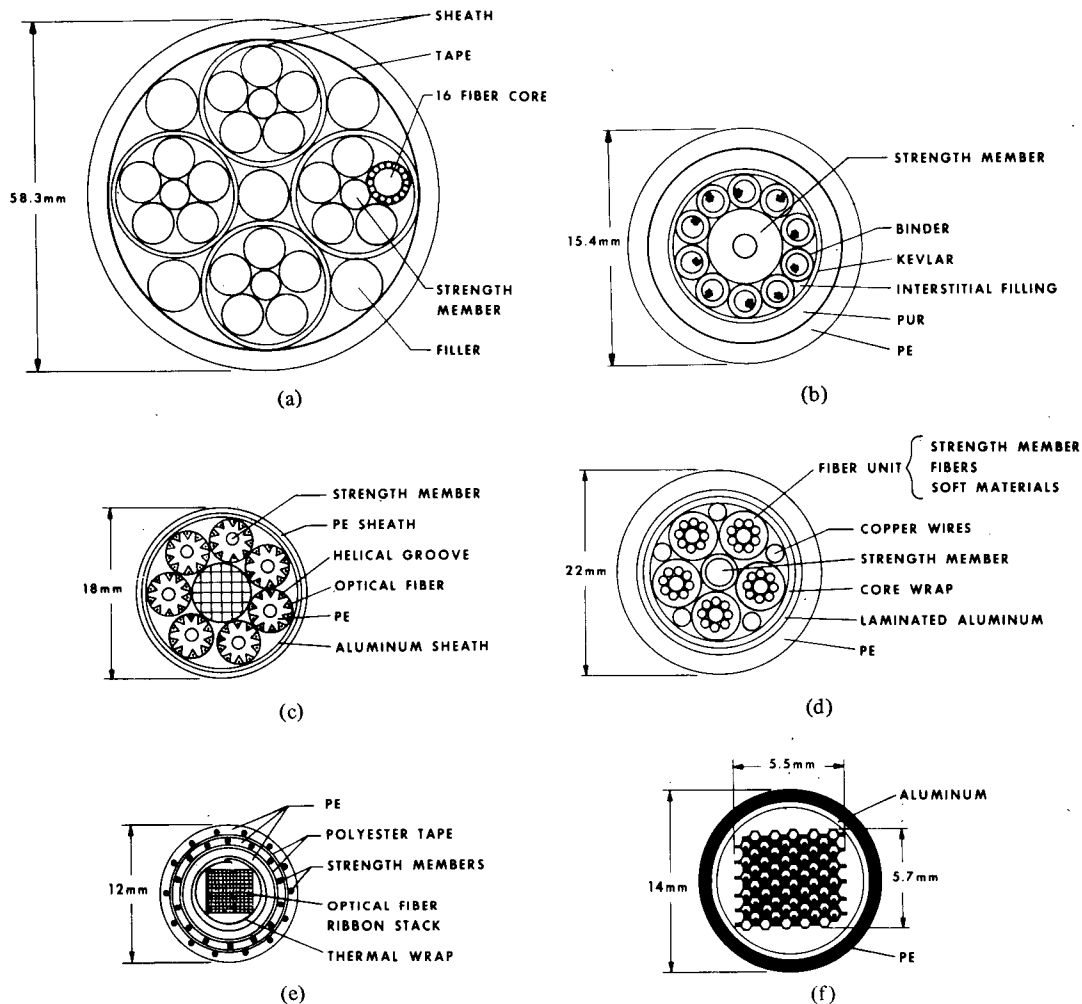


Fig. 4. High density cable designs. (a) 320 fiber-tight stranded. (b) 70 fiber-loose bundles. (c) 70 fiber-loose cylindrical. (d) 40 fiber-tight stranded. (e) 144 fiber-tight ribbon. (f) 50 fiber-loose ribbon.

fibers may be in well defined locations; aiding mass splicing operations.

B. Strength Members

Common materials used to carry tensile loads in optical cables are fiber reinforced plastic (FRP), fiberglass, Kevlar®, and steel. These materials must not only have a high modulus but must be compatible with the other components of the cable, especially during processing of these materials in cable manufacture. Various plastic materials are typically extruded during cable fabrication, and dimensional changes can cause serious problems during manufacture or subsequent environmental temperature extremes if the cable is not properly designed.

Strength members are usually located either in the center of the cable or in the outer sheath. Central strength members offer ease of manufacture, especially in stranded designs since they are the starting element. Advantages of external strength members include greater protection from abrasion and cutting forces, and greater bending stiffness for an equivalent cross-sectional area.

C. Cable Organization

The grooved cylindrical core [8] or stranded unit [11]-[13] can be combined to form large fiber count cables (Fig. 4) necessary for intercity requirements. Loose [9] or tight ribbon [10] core elements are usually stacked to form a cable core. Once units have been combined to form cable cores of desired fiber count, a filling material, moisture barrier, protective sheath, additional strength members, or armoring can be added.

IV. FIBER SPLICES

Following conventional copper media terminology, permanent connections are referred to as splices, whereas connections designed to be disconnected and reconnected many times are referred to as connectors. Typical intercity trunk applications have splices every 0.5-1 km, therefore, each fiber is spliced 8-16 times or more within one repeater span. The large number of splices involved requires splice loss to be kept to a minimum. Most system designs require at least one connector at each end of every span to allow interconnec-

tion capability and test access. Connectors must be able to retain low-loss characteristics through perhaps hundreds of connect/disconnect cycles.

A. Connection Loss

Loss in fiber connections (splices or connectors) is caused by differences in the fibers being connected (intrinsic parameters) and the quality of the connection (extrinsic parameters). Intrinsic parameters include manufacturing variations in fiber outer diameter and core diameters, index profile (α and Δ), and core ellipticity and concentricity. Splice loss due to fiber intrinsic variations typically amounts to 0.1–0.2 dB per splice. Extrinsic parameters include alignment (transverse, angular, and longitudinal) as well as end quality, degree of contamination, and degree of index matching in the connection. Extrinsic loss can be minimized by proper splice design; however, typically in the field environment, 0.1–0.2 dB per splice or more can be attributed to this effect. Fortunately, intrinsic and extrinsic losses combine to give a total splice loss that is less than the algebraic sum.

The overall effect of a given parameter (intrinsic or extrinsic) depends on both the inherent sensitivity to the parameter and the ability to control the parameter. In most cases Δ and core diameter are the most significant intrinsic factors and transverse alignment is the most significant extrinsic factor contributing to splice loss. Splice loss models [14], [15] have been derived to calculate individual and combinations of intrinsic and extrinsic splice parameters, and Fig. 5 shows the resulting losses for the three most important cases. It should be pointed out that, while these three loss mechanisms constitute a major portion of the loss of most high quality field splices, any of the effects, if not properly controlled, can cause high splice loss.

B. Splice Loss Measurements

Multimode fiber splice loss as well as splice loss measurements are very dependent on the power distribution among the propagating modes at the input to the splice. This can be seen in Fig. 5 comparing the uniform power distribution model (equal mode power) to the Gaussian power model (steady-state power distribution). Most investigators attempt to generate a steady-state modal power distribution (one which would exist after several kms of fiber) to measure splice loss. This procedure is difficult, since each fiber has a slightly different steady-state distribution, and is impractical in the field, since splices are usually separated at most by 1 km or so in the metropolitan trunk environment. Perhaps the most accurate way to characterize splices for a given system application is to measure with a typical source and with typical lengths of fibers and splices preceding the splice. The technique of simulating realistic power distributions is particularly important for characterization of connectors which can be located very near sources and other connectors in the field.

Optical time domain reflectometry [16] (OTDR) is one of the most useful techniques for splice loss measurements in the field. OTDR is not a high accuracy technique for an

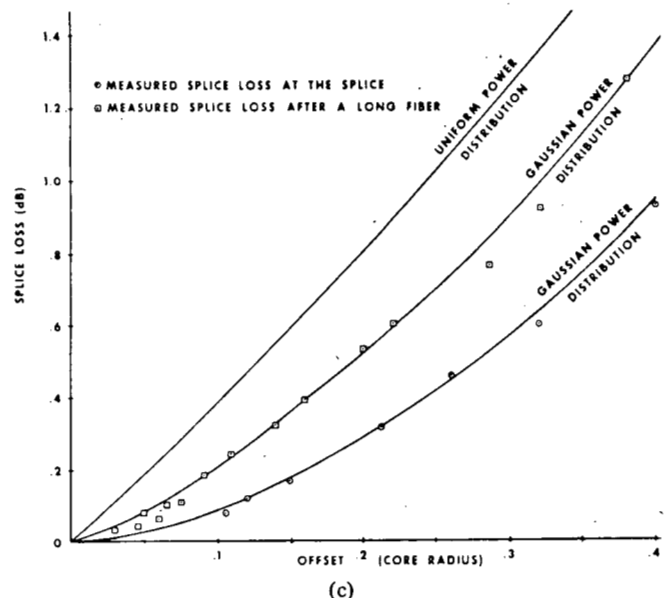
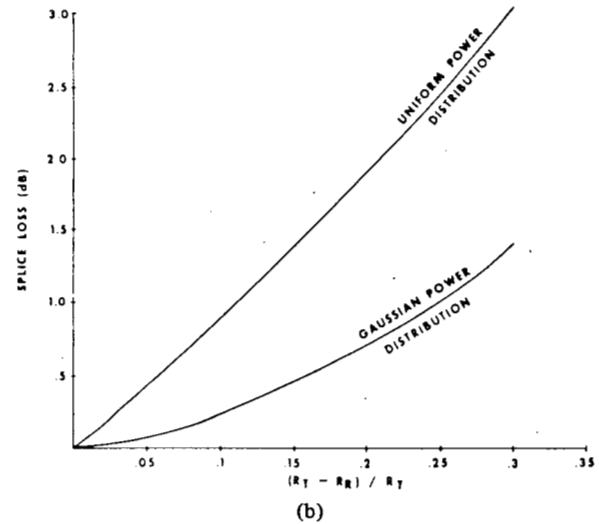
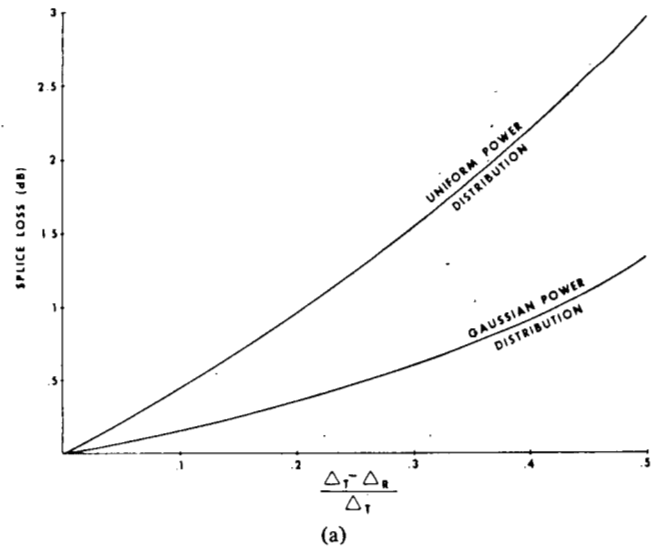


Fig. 5. Splice loss contributions due to (a) Δ mismatch, (b) core radius mismatch, and (c) transverse offset.

individual splice due to an inherent sensitivity to mismatch in backscatter coefficient between the two fibers being spliced; however, this effect averages to zero for a large number of splices. The major advantages of the OTDR are that access to only one end of the fiber is required and measurements can be made quickly and automatically.

C. Splicing Operations

The three fundamental operations involved in fabrication of fiber splices or connectors are: 1) fiber end preparation, 2) alignment, and 3) alignment retention. Methods for end preparation include: 1) scoring and breaking, 2) grinding and polishing, and 3) sawing. Most alignment mechanisms for butt-joint splices use continuous grooves. Since fiber outer diameter variations during manufacture can be held to ± 1 percent or less, and since core eccentricities can be held to a few microns, the continuous groove approach provides adequate core alignment (for multimode fibers) by aligning the fiber outer diameter. Often additional material is applied to the fiber (grooved chips, ferrules, molded plastic, etc.) to increase the size of the structure for ease of handling. Adding material generally increases splice loss, since tolerances on the additional material tends to contribute to fiber misalignment. However, if high precision parts are used, the resulting ease of handling is often worth the slight increase in splice loss. Alignment retention is usually accomplished by springs, adhesives, clamps, etc., or, in the case of fusion, by melting and forming a continuous fiber. Index matching material is often added between the alignment and the retention operation to eliminate Fresnel reflections from the fiber ends.

Environmental variations of splice loss are usually due to a failure in the alignment retention mechanism or in the index matching material. Differential thermal expansion of alignment retention materials can cause misalignment of fibers, and degradation of index matching material with time or temperature can cause absorption or scattering in the splice. Careful material selection and splice design can eliminate these problems [17]. Although fusion splicing would seem to be inherently stable and free from environmental variations, care must be exercised in recoating the fused region, especially if only plastic materials are used. Again, differential expansion in an improperly designed or fabricated fusion recoating can cause loss variations or failure of the joint with time [18].

D. Examples of Single Fiber Splices

Fig. 6 illustrates several techniques for accomplishing single fiber connections. All techniques require fiber end preparation first, then alignment retention by a spring [19], an adhesive [20], [21], or fusion. Typical extrinsic losses for most single fiber techniques are in the 0.02 dB–0.1 dB range for splices made with identical fibers under laboratory conditions. Field results with randomly selected fibers depend on the intrinsic component of loss, and the skill and care of the splicer. Efforts continue to find techniques which do not depend so strongly on the splicer.

Fusion remains the only splicing technique that can result in

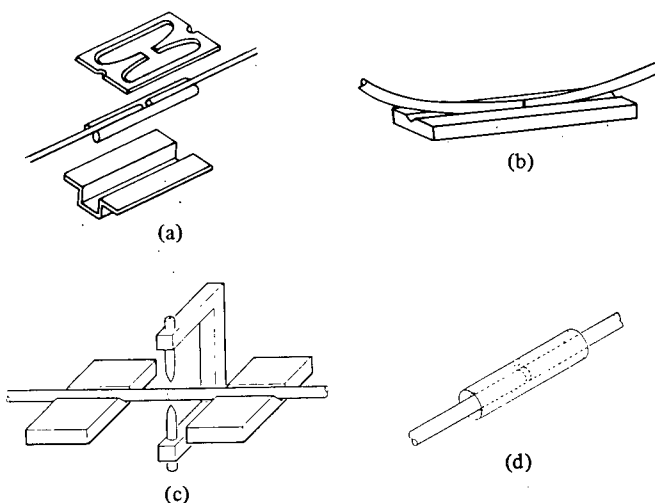


Fig. 6. Single fiber splicing techniques. (a) Springgroove®. (b) V-groove. (c) Fiber fusion. (d) Snug tube.

a splice the same size as the original fiber. Some degree of core deformation must take place during the fusion operation; fusion time and arc or flame temperature can be adjusted to minimize this effect. Partial self-alignment of the fibers takes place under conditions in which the fiber ends become liquid enough to exert a surface tension force. This force aids in the alignment of symmetric fibers but can lead to core misalignment and added splice loss due to fiber asymmetries. Maximizing the strength of fusion joints may require a different set of conditions, and some tradeoff may be required if both high strength and low-loss fusion splices are required.

Fusion splicing has been used extensively in the interoffice trunk applications primarily for small fiber count cables. Splice losses usually average 0.1 dB–0.4 dB. A recent application [12] used individual fiber fusion for a 70 fiber cable and achieved an average splice loss of 0.29 dB for six cable splices. Total splicing time was 48 manhours per cable splice. While the actual fiber splicing time was reported to be only 5–10 min per splice, additional time was required for: 1) initial setup, 2) cable and fiber preparation, 3) splice loss measurement, 4) occasional splice refabrication, 5) spliced fiber organization, and 6) sealing and placement of splice closure. This amounts to $\frac{2}{3}$ manhours per fiber splice.

E. Mass Splicing

Simultaneous splicing of several fibers is known as mass splicing. In the interoffice truck environment, mass splicing can be extremely cost effective without sacrificing performance or environmental stability. Fig. 7 shows two mass splicing techniques currently being used. In the Bell System linear array splice [22], 12 fibers are sandwiched between precision etched silicon chips to form precision linear arrays. Fiber ends are prepared by grinding and polishing, and bridging chips with spring clips are used for alignment retention. As presently practiced, arrays are assembled and inspected in the factory under conditions favorable to high-precision fabrica-

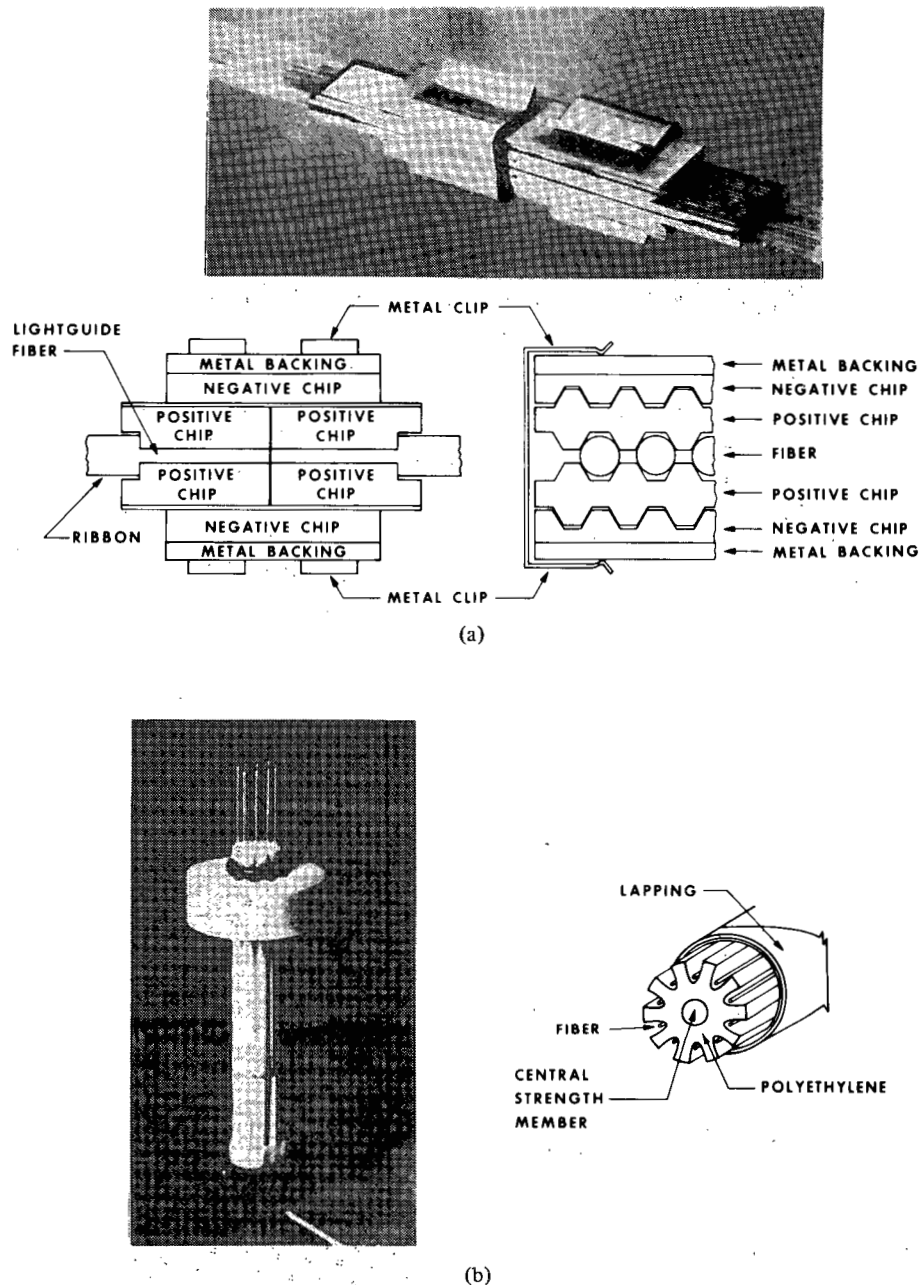


Fig. 7. Examples of mass splicing techniques. (a) Linear array.
(b) Grooved cylinder.

tion. In the field, the arrays are clipped together, index matched with a silicone gel and measured by OTDR. Splicing time is approximately one min per fiber. Since both ends of the cable are preconnectorized before installation, route surveys are necessary to accurately determine cable length. Some excess cable storage (a few meters) is generally necessary. Assembly of the linear arrays in the field is recommended as a repair technique only; however, field assembly does not require handling individual fibers.

A grooved cylindrical structure [23] is used in the second mass splicing technique illustrated in Fig. 7. Fibers are placed in the grooves all at once without handling individual fibers, cemented in place, and end preparation accomplished by sawing. Mating grooved cylinder segments are used for each splice to minimize differences in alignment structures. Average

splice loss is reported to be 0.3 dB/splice. Since this splice is totally assembled in the field, an accurate route survey is not required.

V. BELL SYSTEM RESULTS

As of December 1, 1982, more than 150 000 km of cabled fibers have been installed by operating personnel in 27 different Bell System standard lightwave transmission systems (designated FT3 and FT3C operating at 45 and 90 Mbits/s, respectively). Over 228 000 fiber splices using linear arrays have been connected in the field (usually in manholes) with an average loss below 0.2 dB. No instances of problems resulting in the permanent loss of a single fiber path due to installation breaks or high loss in splicing have occurred. This represents an incredibly good record for such a new technology. Time required for

mass splicing and testing is currently less than one min per fiber. (Additional time for travel-to-site, manhole pumpout, etc., is not included.) Typical examples of current splicing times are: 1) two eight ribbon (96 fiber) cable splices in different manholes completed in 11 hours total by one two-man splicing crew, and 2) a 12 ribbon (144 fiber) and an eight ribbon cable splice in the same manhole completed in nine hours total by one two-man splicing crew. An additional person is required to make OTDR measurements as splices are completed. These total splicing times of less than seven man-min per splice, made possible by a coordinated cable and splice design for mass splicing, have saved many thousand man-hours in field splicing. Approximately 200 arrays have been fabricated in the field [24] by telephone company craft personnel due to array defects or installation and splicing accidents. These splices have an average loss of 0.23 dB indicating that field array fabrication is also a viable alternative to individual fiber splicing.

The Bell System, along with many other companies, has had extremely favorable experiences with optical fiber trunk systems proving that, although unique challenges exist in optical cable and splicing design, these challenges can and are being met. Approximately 22 percent of the Bell System fiber installed thus far has been for an FT3C long-haul application in the Northeast Corridor, a network 776 miles long from Moseley, VA, to Cambridge, MA. A 500 mile FT3C system linking Los Angeles and San Francisco is also under construction.

VI. FUTURE CABLE AND SPLICING REQUIREMENTS

Applications of single-mode fibers in the long-haul and multimode fibers in the distribution portions of the telephone network have lagged trunk applications primarily because early components, devices, and multimode fiber were well suited to trunk use. As long wavelength devices and single-mode fiber and components become available, long-haul fiber systems are being designed and tested. British Telecom has an operational single-mode fiber link (27 km) and plans many additional long-haul routes. Repeater spans of 20–50 km and bandwidths greater than 1 GHz appear feasible with single-mode fibers at 1.3 μm and 1.55 μm . These extremely low-loss characteristics must be preserved by cable and splice designs that contribute little or no added loss. Splicing requirements for single-mode fibers are fundamentally different than for multimode fibers. Fig. 8 shows the increased sensitivity to offset due to reduced core diameter. Also, accurate core centering relative to fiber outer diameter may not be practical in large scale manufacture; therefore, active alignment during splicing most likely will be required to achieve minimum splice loss. These considerations, along with the need for the absolute lowest loss, rule out mass splicing at the outset. Fortunately, long-haul splices will be fewer in number due to longer cable lengths (perhaps 1–3 km) and fewer fibers per cable (perhaps 10–40). Since each fiber carries an even larger number of circuits, a larger cost per fiber splice is economically viable.

Fibers installed in the distribution portion of the network have the potential of providing a wide bandwidth path directly to the customer thereby enabling new services to be offered. Several countries have installed experimental local distribu-

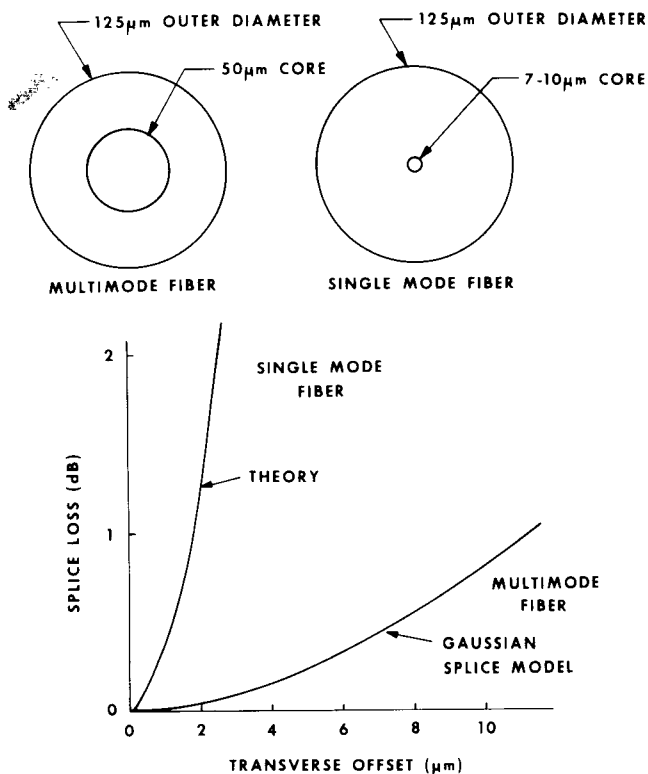


Fig. 8. Comparison of multimode and single-mode geometry and splice loss due to offset.

tion systems to generate needed technology and test customer acceptance of new services. Cable requirements for the distribution portion of the network include: 1) large numbers of fibers in each cable, 2) varied environment—aerial, buried, underground, and 3) increased emphasis on low cost. In addition, the distribution plant may be highly tapered, that is, many communication paths needed near the central office or remote terminal and fewer near the most remote customers. This tapering requires many splices for dropping off fibers, branching, and transitions to different fiber count cables. Splice loss requirements will probably be eased since distribution lengths are typically only a few kilometers. Splicing time which translates directly to cost becomes most important in the distribution plant, making mass splicing of large distribution cables attractive. The requirement for increased connection flexibility, however, especially for small fiber count distribution cables, often dictates individual fiber splicing.

Long-haul and distribution systems will offer different challenges to the optical fiber cable and splice designer. These challenges will undoubtedly be met, allowing a host of new and as yet undetermined wide-band services to be provided to business and residential customers.

REFERENCES

- [1] W. B. Gardner, "Fundamental characteristics of optical fibers," *Telecommun. J.*, vol. 48, Nov. 1981.
- [2] G. LeNoane, "Optical fiber cables," *Telecommun. J.*, vol. 48, Nov. 1981.
- [3] M. I. Schwartz, P. F. Gagen, and M. R. Santana, "Fiber cable design and characterization," *Proc. IEEE*, vol. 68, Oct. 1980.
- [4] B. R. Eichenbaum, M. R. Santana, L. D. Tate, and R. Sobin, "Design and performance of a filled high-fiber-count, multimode optical cable," presented at the 31st Int. Wire Cable Symp., Nov. 1982.

- [5] D. Kalish, B. K. Tariyal, and H. C. Chandan, "Effect of moisture on the strength of fibers," presented at the 27th Int. Wire Cable Symp., Nov. 1978.
- [6] W. B. Gardner, "Microbending loss in optical fibers," *Bell Syst. Tech. J.*, vol. 54, Feb. 1975.
- [7] U. Oestreich, "Fiber optical cable for aerial application," presented at the 29th Int. Wire Cable Symp., Nov. 1980.
- [8] M. deVecchis, G. LeNoane, and J. P. Hulin, "Experimental results of cylindrical V-grooved structure optical cables laid, in ducts and spliced," presented at the 4th Euro. Conf. Opt. Commun., Genova, Italy, Sept. 1978.
- [9] R. Jocteur and M. Carratt, "A new fiber ribbon cable," presented at the 6th Euro. Conf. Opt. Commun., York, England, Sept. 1980.
- [10] M. J. Saunders and W. L. Parham, "Adhesive sandwich optical fiber ribbons," *Bell Syst. Tech. J.*, vol. 56, July/Aug. 1977.
- [11] W. Schmidt, W. Auer, K. Kimmich, I. Rigel, and U. Zwick, "320-core optical fiber cable," presented at the 6th Euro. Conf. Opt. Commun., York, England, Sept. 1980.
- [12] R. A. Greulich and R. K. Lux, "PNB installs underwater FO link—A 'landmark' job," *Telephony*, Feb. 22, 1982.
- [13] R. Yamauchi, K. Inada, Y. Sugawara, and T. Kobayashi, "Elongations of optical fibers in an optical cable under installation," presented at the 4th Euro. Conf. Opt. Commun., Genova, Italy, Sept. 1978.
- [14] C. M. Miller and S. C. Mettler, "A loss model for parabolic-profile fiber splices," *Bell Syst. Tech. J.*, vol. 47, Nov. 1978.
- [15] S. C. Mettler, "A general characterization of splice loss for multimode optical fibers," *Bell Syst. Tech. J.*, vol. 58, Dec. 1979.
- [16] P. Matthyse and C. M. DeBlok, "Field measurement of splice loss applying the backscatter method," *Electron. Lett.*, vol. 15, Nov. 22, 1979.
- [17] S. C. Mettler and M. R. Gotthardt, "Index-of-refraction matching materials for optical fiber splicing," presented at the 3rd Int. Conf. Integrated Opt., Opt. Fiber Commun., Apr. 1981.
- [18] S. Stueflotten, "Protection of optical fiber arc fusion splices," *J. Opt. Commun.*, vol. 3, p. 1, 1982.
- [19] G. Cocita, B. Costa, S. Longoni, L. Michetti, L. Silvestri, D. Tribone, and F. Tosco, "COS 2 experiment in Turin: Field tests on an optical cable in ducts," *IEEE Trans. Commun.*, July 1978.
- [20] D. Kunze, H. J. Krimmling, H. Liertz, and E. Bachel, "Jointing technique for optical cables," presented at the 2nd Euro. Conf. Opt. Fiber Commun., Paris, France, Sept. 1976.
- [21] M. Murata, "Splicing of optical fiber cable," presented at the 1st Euro. Conf. Opt. Commun., London, England, Sept. 1975.
- [22] C. Miller, "Fiber optical array splicing with etched silicon chips," *Bell Syst. Tech. J.*, vol. 57, Jan. 1978.
- [23] A. Bouvard and J. P. Hulin, "Results on field operation of a mass splicing process," presented at the 7th Euro. Conf. Opt. Commun., Copenhagen, Denmark, Sept. 1981.
- [24] H. J. Friedrichsen and P. F. Gagen, "Optical cable field repair using array fabrication," presented at the 7th Euro. Conf. Opt. Commun., Copenhagen, Denmark, Sept. 1981.



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